



TME 6202- Technology and Operations Management

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Final Project Report

Group 4

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Research Statement

How can AMCORP improve cross-site resource traceability and utilization across concurrent EPC projects.

Company and Operational Context

AMCORP Engineering & Construction (Pvt.) Ltd., established in 1965, has Pakistan's highest contractor category (PEC Category CA) and is certified ISO 9001, 45001 and 14001. Around 155 full-time resources are engaged in multiple EPC projects.

At present; a 28-acre port expansion for DP World at QICT Karachi, Pakistan's tallest tower at KGTML, an 18-storey commercial complex for Allied Bank at Lahore, remote O&G works for PPL at Sirani Block, electromechanical works for Siemens at Gharo, and infrastructure for EMAAR; all are managed from a single warehouse in Karachi. Current ERP is a ledger with no GPS linking, no RFID info feeding, no offline capability and no multi-site view. Reporting is done every week but with 8-10 days delays and stock ledgers are held in hardcover books, copied by hand.

Research Question and Problem Statement

How can AMCORP improve cross-site resource traceability and utilization across concurrent EPC projects to eliminate procurement waste, reduce idle assets and idle workers, and establish real-time operational control? The biggest financial losses have three sources of pain:

First, all purchased items are already stocked elsewhere at an AMCORP site; UPVC pipes, cable coils, scaffolding purchased that are sitting somewhere, unknown to the other projects. **Second**, idled labour waiting for equipment is a hidden cost that is the highest. **Third**, emergency procurement at 30-40% mark-up because contracting and CEO approval of vendor quotations takes 3-7 days per purchase order. Every project also ends with excess materials that cannot be sold because they are not labelled properly in the system on time.

KPIs: Before and After

Seven major KPIs were identified from the C-level interviews:

- 1) Worker idle cost per incident is huge; up to Rs. 3,70,000 confirmed in one case, targeting near zero.
- 2) Cross-site inventory query lag is eight to ten days for sites, targeting under 24hrs.
- 3) Emergency procurement needs premium payment for 30-40% percent of the cases, targeting below 10%.
- 4) Duplicate procurement rate is 15-25% percent of all purchase orders, targeting below 5%.
- 5) Indirect item inventory accuracy is currently at zero real-time visibility, targeting a live RFID register.
- 6) Project margin variance, anchored in the DP World versus PPL outcome, targeting consistent positive margins across all projects.
- 7) CEO approval cycle on every purchase order, regardless of size, targeting under 24hrs through a pre-loaded vendor rate library.

Process Map: Before and After

Today: Site sees a need; unable to check other sites the need is raised by phone or paper; worker idle time begins invisibly; four to five vendor quotes are sourced manually; a three to seven day approval process commences; material arrives and a paper GRN (Goods Received Note) is completed at site; a weekly report goes to HQ 5-10 days later; surplus at project end is scrapped. Each hand-off is human memory dependent.

After implementation: Site files demand with online or offline-first mobile app, no internet required (critical for PPL Sirani and Gharo sites), the system checks for availability across site locations in seconds; if found, COO approves single-tap transfer; if external purchase is needed, the rate library generates a draft purchase order (PO) and CEO approves within 24 hours; delivery is automatically recorded via an RFID gate reader; COO's interactive Power BI dashboard shows all sites live with a 15 minute maximum lag; surplus alert triggered 30 days before project closeup, recycles 80% of what would otherwise be scrap.

Proposed Solution

The architecture is an I4.0 project in 3 phases for a total of PKR14,483,420 over two years.

Phase 1 (PKR 435,000, Weeks 1-12) is a free process improvement: COO mandate that all sites check stock across all sites before making a purchase over PKR 50,000, virtual ledger across sites, report template, and KPI ownership by site manager; expected to reduce double procurement errors to less than twelve percent by hardware deployment.

Phase 2 (PKR 4,948,000, Months 3-9) is physical tracking: UHF RFID at all site stores and tags on all items above PKR 10,000 value, GPS telematics on 20 heavy machines via Teltonika FMB920s, QR checkout for the stores instead of paper GRNs, and an offline-first Progressive Web App, the most important component (because PPL Sirani and Ghara have no internet, and anything requiring connectivity fails on first use).

Phase 3 (PKR 1,891,000, Months 6-12) adds LoRaWAN vibration sensors, fuel gauges and temperature sensors for preventive maintenance and theft protection.

Phase 4 (PKR 7,209,420, Months 12-24) is integration: Odoo Enterprise ERP to link procurement, inventory, maintenance and finance across sites; Azure IoT Hub to normalise all hardware data streams; Fiix CMMS to create predictive maintenance work orders automatically from sensor thresholds; a built-in vendor rate library for the top fifty vendors and top two hundred SKUs; and a single Power BI COO dashboard showing every site, 24/7, replacing manual weekly reports.

Literature Review

Khan, Ray and Karmakar (2024, Automation in Construction) showed 75% of construction companies have the same lack of visibility as AMCORP and that 99.8% RFID accuracy is achievable when using metal in construction. Abdelmoti et al. (2025, Buildings) found 46% of industry still relies on Excel and 41.2% blame material waste on lack of coordination between site and head office, thus statistically confirming each pain point. Goodrum, Haas and Glover (University of Kentucky) found 35% less labour idle time by using material tracking technology, academic confirmation of the same KPI. Cheng and Teizer (Georgia Tech) validated 25-35% of material waste reduction with IoT-based material tracking and LoRaWAN to be an option for tracking in remote sites (no cell signal), addressing PPL Sirani.

Global and Local Benchmarks

Several EPC companies have achieved results with reported work improvements. We will discuss 3 in our research: Bechtel (Australia) used RFID in a 50 site supply chain to drive steel delivery from 45 to 22 days, Descon Engineering (Pakistan) worked by moving from multiple sources to a stored "single version of truth" for SAP EC&O, which provides centralised procurement processes, material tracking at project level and financial control and Habib Construction Services (Pakistan) overcame this problem on a financial level only when it expanded, by adopting ERP (by BE Tech) software indirectly linking to the problems of AMCORP.

Business Impact for AMCORP

The estimated annual saving is PKR 8-25 million in five areas: no duplicate procurement (PKR 3-8M), 25% rental reduction for equipment hire (PKR 1.5-5M), no emergency premium for equipment (PKR 1-4M), recover worker idle time (PKR 1-4M), and recover material left over (PKR 0.5-4M per project at 80% recovery). The payback period on the investment (PKR 14,483,420 over 24 months) is 18-24 months, conservatively. Phase 1 alone (PKR 435,000) is projected to save PKR 3 to 6 million a year before any equipment is purchased.

Every technological innovation comes with its uninvited risk: a high number of employees at the frontier who are less tech savvy and known to older ways, would make a task of cultural change along with the technical change a challenge. C-levels defined the problems themselves and helped us quantify them and agreed that they are achievable but the only variable is when.

References

1. Abdelmoti, A. M., Shafiq, M. T., Rauf, A., & Khalfan, M. M. A. (2025). *Challenges in Asset Management and Digital Twins: Industry Insights*. *Buildings*, 15(11), 1809.
2. Bechtel Corporation. (2014, April 22). *Bechtel takes top honor for innovative use of RFID technology* [Press release].
<https://www.bechtel.com/press-releases/bechtel-takes-top-honor-for-innovative-use-of-rfid-technology>
3. Cheng, T., & Teizer, J. (2013). *Real-time resource location data collection and visualization technology for construction safety and activity monitoring applications*. *Automation in construction*, 34, 3-15.
4. Goodrum, P. M., Haas, C. T., & Glover, R. W. (2002). *The divergence in aggregate and activity estimates of US construction productivity*. *Construction management & economics*, 20(5), 415-423.
5. Khan, S. I., Ray, B. R., & Karmakar, N. C. (2024). *RFID localization in construction with IoT and security integration*. *Automation in Construction*, 159, 105249.
6. LNG Industry. (2014, April 9). *Bechtel wins award for technology on LNG project*.
https://www.lngindustry.com/liquid-natural-gas/09042014/bechtel_wins_lng_technology_award_410/
7. PACRA. (2024, March 15). *Descon Engineering Limited — credit rating report* (Report No. RR_2058_12682). https://www.pacra.com/summary_report/RR_2058_12682_15-Mar-24.pdf
8. SUSE. (2013). *Descon Engineering Limited maintains global lead with SUSE Linux Enterprise Server for SAP applications* [Press release].
<https://www.suse.com/news/descon-engineering-limited-maintains-global-lead-with-suse-linux-enterprise-server-for-sap-applications/>
9. PACRA. (2024, October 11). *Habib Construction Services Limited — credit rating report* (Report No. RR_1410_13659). https://www.pacra.com/summary_report/RR_1410_13659_11-Oct-24.pdf